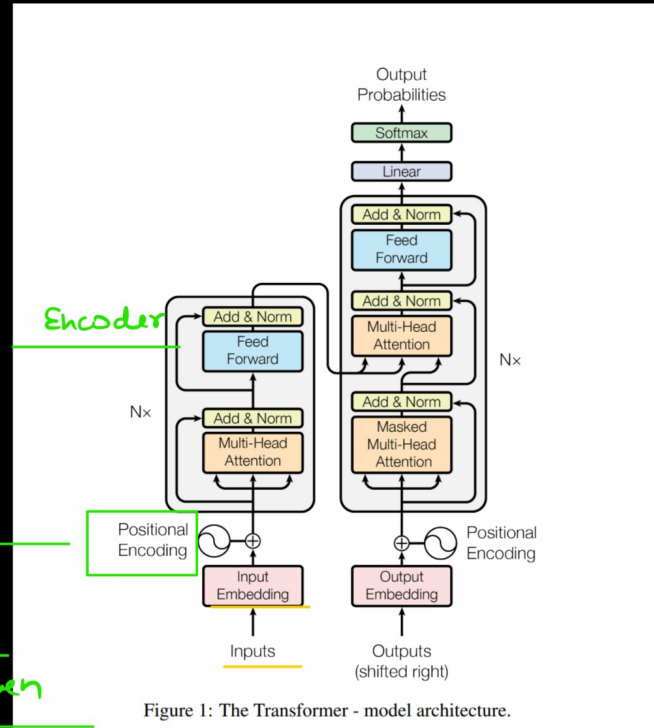
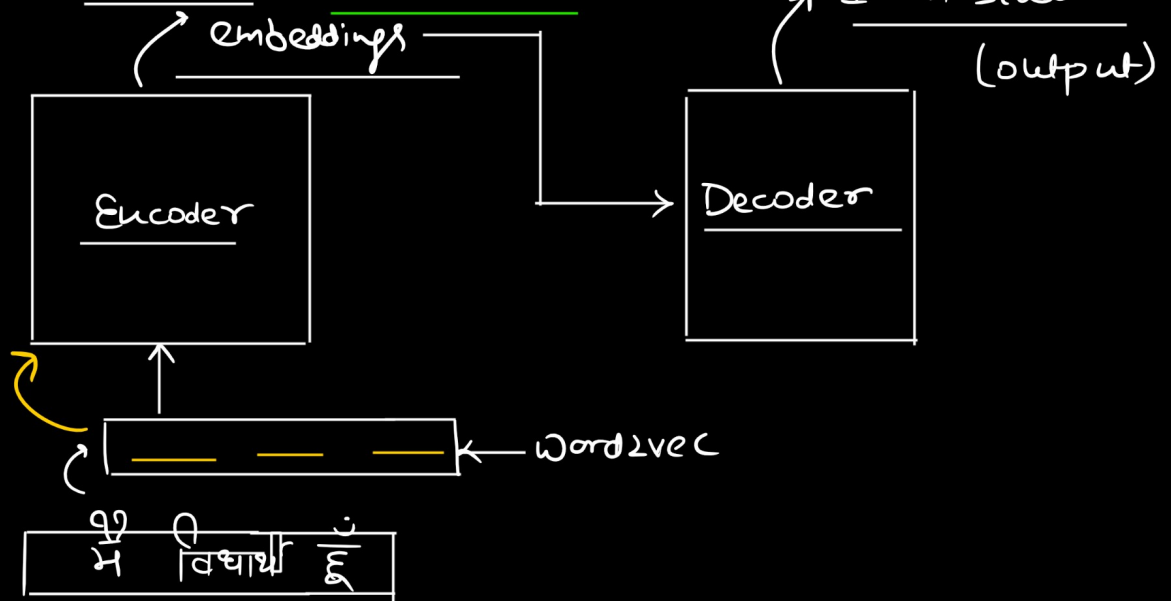


Transformer Architecture

I love AI
 ↓ ↓ ↓
 0 1 2

understand
 the sequence
 of the given

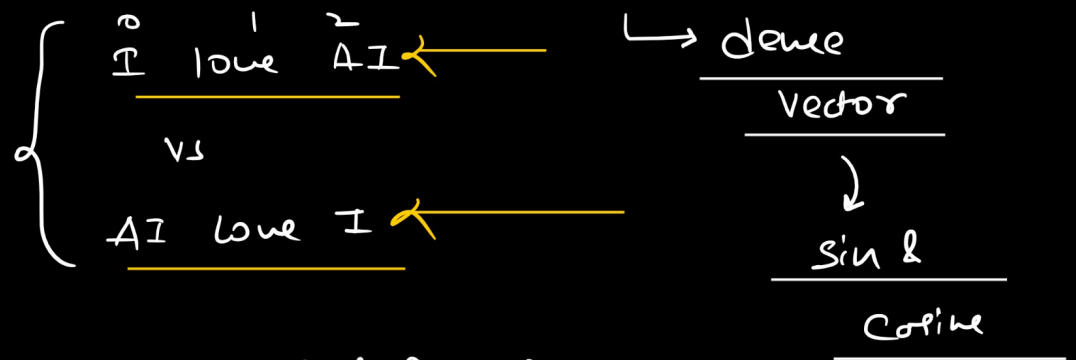
contextual words



① It process all the words simultaneously

② Positional encodings are added to
 the embeddings

Word embeddings + Position information $\Rightarrow$ position aware representations



Self Attention

$\begin{array}{ccc} 1 & 2 & 3 \\ [I & \text{love} & AI] \end{array}$



10 dimensional vector α — Word embedding

10 dimensional
Vector

α — Positional
 encoding

$\left\{ \begin{array}{l} [0.84, \\ 0.34, \\ 0.10, \\ 0.98, \\ 0.01, \\ 0.999, \\ 0.001, \\ 0.2, \\ 0.3, \\ 1.0] \end{array} \right.$

money bank grows

$$\begin{cases} \underline{e_{\text{money}}} \Rightarrow 0.7e_{\text{money}} + 0.2e_{\text{bank}} + 0.1e_{\text{grows}} \\ \underline{e_{\text{bank}}} \Rightarrow 0.25e_{\text{money}} + 0.7e_{\text{bank}} + 0.05e_{\text{grows}} \\ \underline{e_{\text{grows}}} \Rightarrow 0.1e_{\text{money}} + 0.2e_{\text{bank}} + 0.7e_{\text{grows}} \end{cases}$$

river bank flows

$$\begin{cases} \underline{e_{\text{river}}} \Rightarrow 0.8e_{\text{river}} + 0.15e_{\text{bank}} + 0.05e_{\text{flows}} \\ \underline{e_{\text{bank}}} = 0.2e_{\text{river}} + 0.78e_{\text{bank}} + 0.02e_{\text{flows}} \\ \underline{e_{\text{flows}}} = 0.4e_{\text{river}} + 0.01e_{\text{bank}} + 0.59e_{\text{flows}} \end{cases}$$

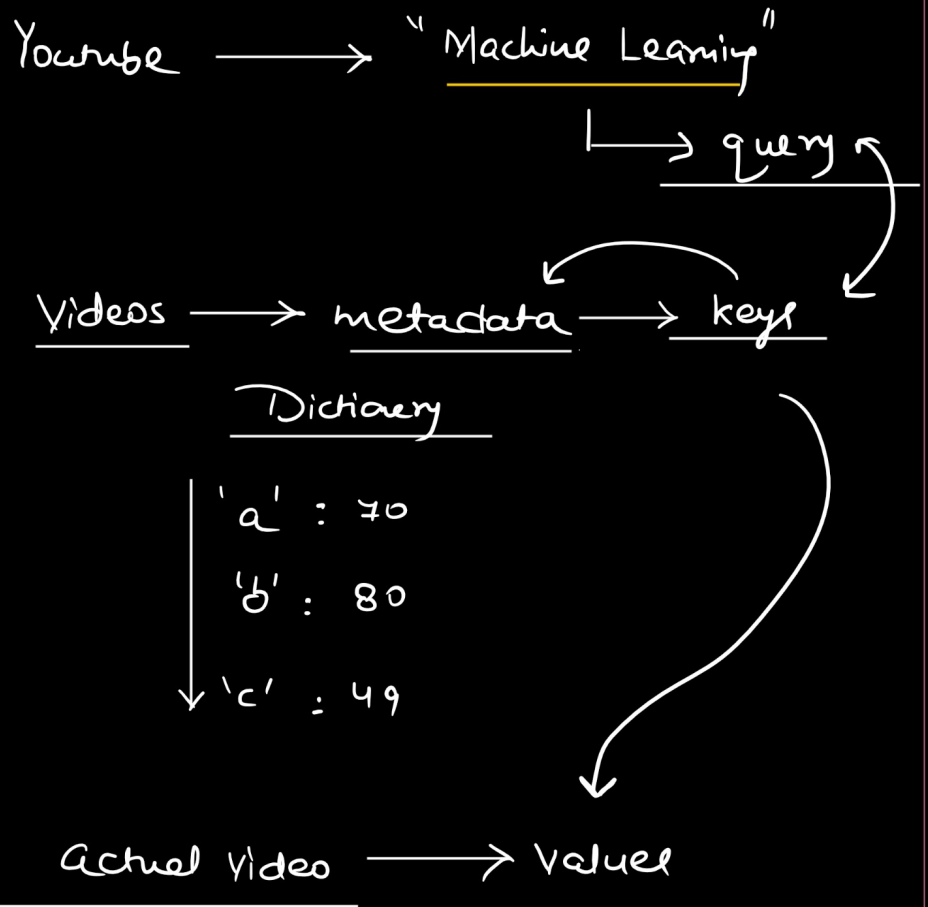
Attention in transformers is the mechanism that allows a model to process sequences (like text) by selectively focusing on the most relevant parts of the input. It determines how much weight or "attention" one token (like a word) should pay to every other token in the same sequence

self-attention mechanism

The cat sat on the mat because it was too
tired.

cat & not
mat

Query, Key, Value



Every Word

$\downarrow$

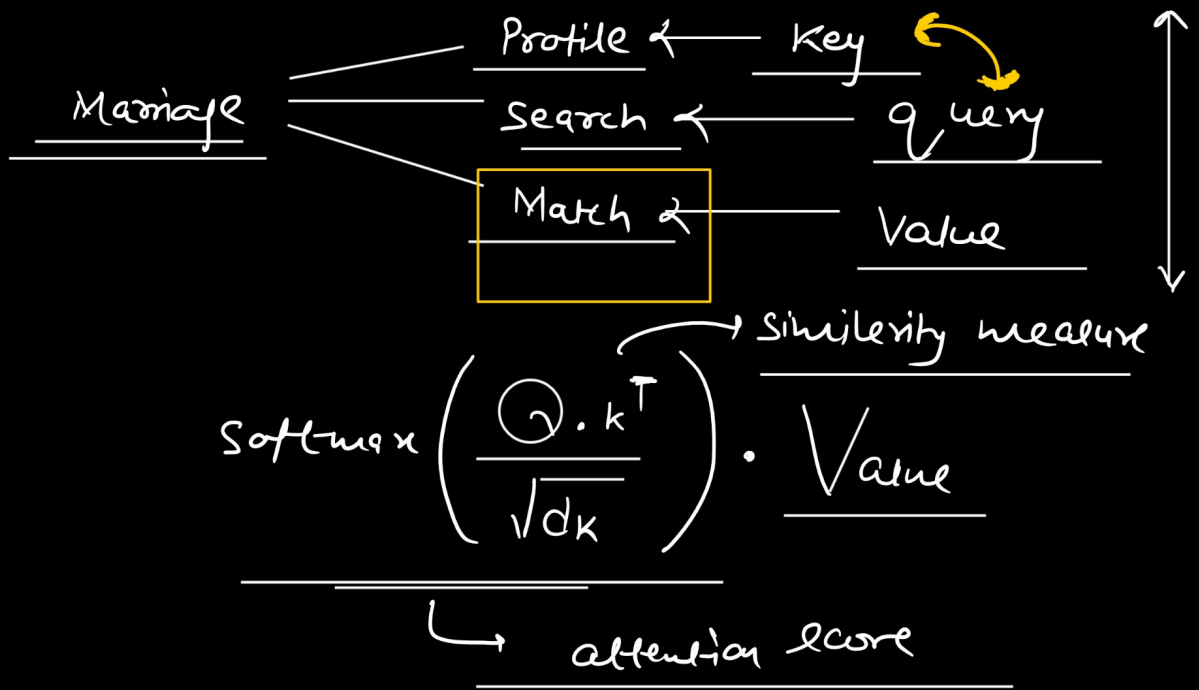
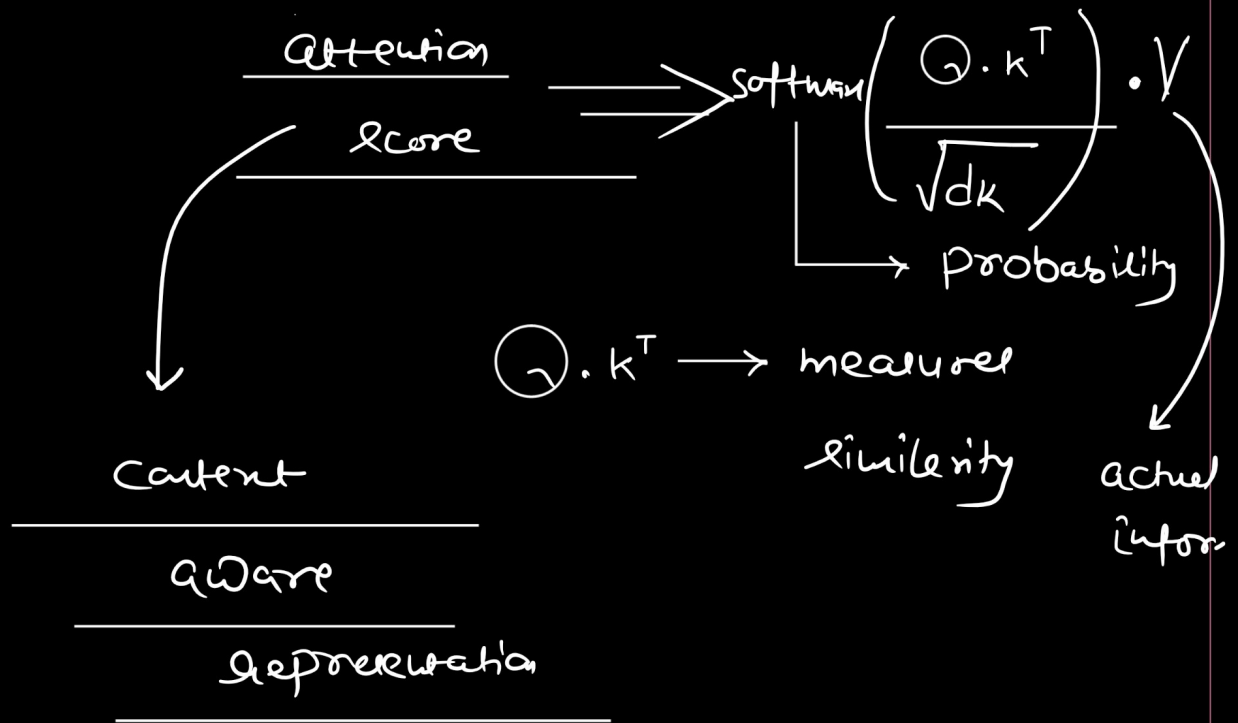
creates query, key, value

for each token (w) $\longrightarrow$ input

$\downarrow$ embedding

$$Q = X W^Q$$
$$K = X W^K$$
$$V = X W^V$$

$W^Q, W^K, W^V \longrightarrow$ learnable parameters



Multihed attention mechanisms

- 1st $\rightarrow$ Grammar
- 2nd $\rightarrow$ semantic meaning
- 3rd $\rightarrow$ long term relationships

Decoder

↳ Masked multi head attention

unsupervised

training

I love Machine Learning
 x_1 x_2 y

$P(y | x_1, x_2) \Rightarrow$ _____

Language model

Evolution of transformer

BERT $\longrightarrow$ Encoder only $\longrightarrow$ Classification

GPT $\longrightarrow$ Decoder only

T5 $\longrightarrow$ Encoder + Decoder

↳ translation

Summarization

generation

multi-task NLP